



TVET MARA LUMUT

DIPLOMA IN COMPUTER TECHNOLOGY (BIG DATA)

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SEMESTER 3

FIBER OPTIC INTERMEDIATE – DFV 31094

PROJECT REPORT

STUDENTS NAME	MATRIX NUMBER
AIMAN BIN RAZALI	2501085
DATU REZWAN'NUR BIN DATU MARWAN	2501086
HAIRUNNISHA BINTI ATTAN	2501087

PREPARED FOR:

AHMAD AL-ZAFRI BIN ANASSUZASTRI

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INTRODUCTION

This project was carried out to install and connect fiber optic cables from the Fiber Distribution Cabinet (FDC) to Cabin 3 at the project site. The main purpose of this project is to provide a stable and high speed communication link between network equipment located in the cabins. The scope of work includes planning the cable route, designing the layout, and installing major components such as manholes, closures, poles, and the Optical Network Unit (ONU) to ensure the fiber optic system functions effectively.

The installation process was conducted in several stages, starting from site surveying, cable length measurement, and routing, followed by connection and final testing. Each step was performed based on the project plan to ensure that the cables were properly and safely installed according to technical standards. The work also required accuracy in measurement and technical skill in fiber splicing to prevent signal loss and maintain optimal network performance.

In addition to the technical aspects, this project also emphasized the importance of safety at the worksite. All team members were required to use Personal Protective Equipment (PPE) such as safety helmets, reflective vests, gloves, and safety boots to minimize the risk of accidents during installation. Through this project, students gained not only technical knowledge in fiber optic systems but also real world experience in fieldwork management and the importance of teamwork to ensure project success.

OBJECTIVE

The main objective of this project is to carry out the installation and connection of fiber optic cables from the Fiber Distribution Cabinet (FDC) to Cabin 3. This project was conducted as part of the Fiber Optic Intermediate practical assessment, where students are required to demonstrate their understanding and technical skills in planning, installing, and testing a fiber optic network system. The entire process was carried out based on the given project plan and technical specifications to ensure that the installation meets the required standards. The objectives of this project are as follows:

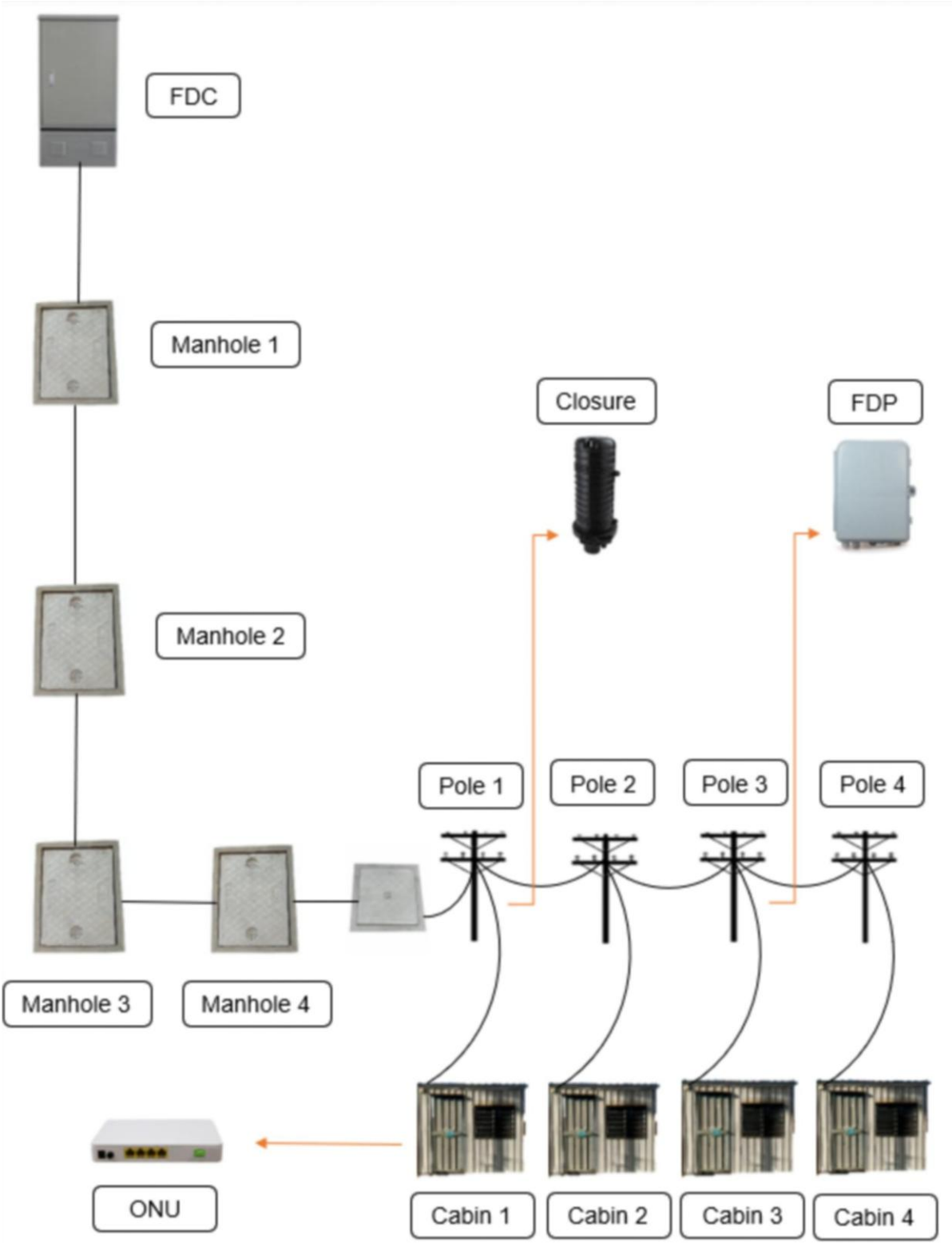
- To install and connect fiber optic cables from the FDC to Cabin 3 according to the project plan and technical specifications.
- To ensure all fiber splicing and cable routing are completed accurately to achieve stable and efficient data transmission between the designated network points.
- To ensure the entire installation process complies with safety standards, with all team members using appropriate Personal Protective Equipment (PPE) to prevent workplace accidents.
- To provide students with hands on experience in real world fiber optic installation work, including handling tools and equipment, performing cable measurements, splicing, and testing.
- To apply teamwork and problem solving skills during the fieldwork process.
- To produce a fully functional fiber optic connection that supports reliable network communication within Cabin 3.

PROJECT SITE



The project site is situated within the campus area of Institut Kemahiran MARA (IKM) Lumut, specifically covering the cable route from the Fiber Distribution Cabinet (FDC) to Cabin 3. Cabin 3 serves as one of the designated termination points within the internal network infrastructure, responsible for receiving and distributing the fiber optic signal to the connected premises. The installation route spans across various infrastructures including underground manholes, utility poles, and fiber closures, all of which were carefully planned and surveyed prior to the commencement of the installation work.

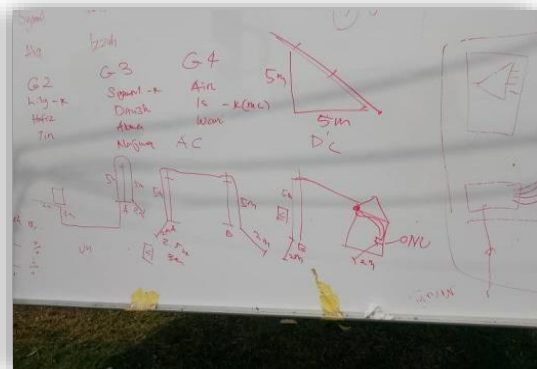
PROJECT PLANNING DESIGN



PREPARATION BEFORE INSTALLATION

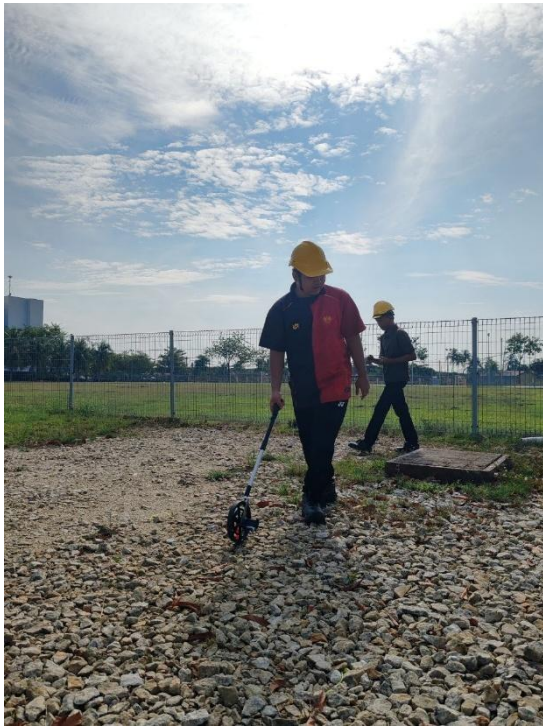
PHASE 1: PLANNING AND SITE SURVEY

Prior to commencing any installation work, the team conducted a comprehensive site survey and safety assessment to evaluate the overall readiness of the project area. The survey covered several key aspects including the condition of the cable path, the state of each manhole along the route, the readiness of the utility poles, and the exact location of the project site. This was done to identify any potential issues that could disrupt the installation process before work officially began.



PHASE 2: MEASUREMENT AND CABLE PREPARATION

In this phase, a measurement wheel was used to measure the exact cable route from the FDC to the ONU. The measurements covered the aerial cable, underground cable, and drop cable sections. Based on the measurements taken, the required cable lengths were cut and prepared accordingly to minimize wastage and ensure accurate installation.



PHASE 3: CABLE STOCK COLLECTION AND CUTTING PREPARATION

Upon completing the measurement phase, the team collected the required fiber optic cables from the store, including the underground cable, aerial cable, and drop cable based on the project specifications. Before bringing the cables to the installation site, the following verification checks were carried out on each cable drum:

- Cable type and core count were verified to match the project specifications.
- Each drum was inspected for any visible damage on the outer sheath.
- The drum number and available cable length were recorded for documentation purposes.



PHASE 4: SAFETY ASSESSMENT AND PPE COMPLIANCE

Before any work commenced on site, a hazard identification and safety assessment was carried out. All team members were briefed on the safety procedures and were required to wear complete Personal Protective Equipment (PPE) including safety helmet, reflective vest, gloves, and safety boots throughout the entire project duration.

➤ Splicing Equipment



Fusion splicer



Fiber cleaver



Fiber stripper



Fiber Cleaning Alcohol



Jacket stripper

➤ Personal Protective Equipment (PPE)

The safety and well-being of every team member on site was given the utmost priority throughout the entire installation process. Before any work commenced, all crew members were required to wear complete Personal Protective Equipment (PPE) to minimize the risk of injury and comply with the required safety standards. The PPE used during this project included:



Safety glove



Reflective Vest



PVC safety boots



Safety Helmet



Safety Shoes

PHASE 5: UNDERGROUND WORK SAFETY PROTOCOL

Before any worker enters the manhole, a series of safety procedures were carried out using the appropriate equipment to ensure a safe working environment. The blower machine was operated first to ventilate the confined space and remove any toxic or hazardous gases that may have accumulated inside. A water suction machine was then used to pump out any stagnant water present at the bottom of the manhole. Once the manhole was cleared and safe for entry, a manhole cover lifter was used to safely remove the heavy manhole cover without risking injury to the workers.



Crowbar



Manhole cover (keys)



Generator



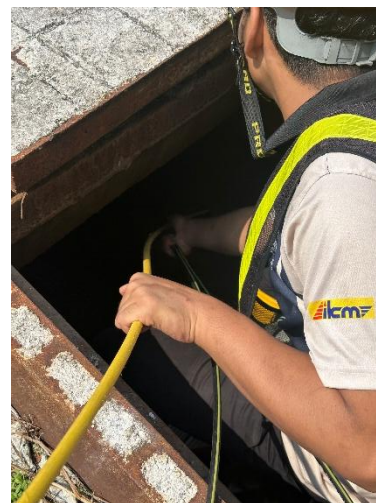
Suction machine

PROJECT INSTALLATION

1. Underground Cable Installation

The underground cable installation was the first stage carried out in the cable installation phase. The following steps were performed during this process:

- Manhole covers were opened along the designated cable route using a manhole cover lifter.
- The water suction machine was used to clear any stagnant water accumulated inside the manhole.
- Once the manhole was confirmed safe, the team began pulling the underground cable through the underground duct from one manhole to the next.
- The cable was carefully guided through the duct to avoid any kinking or damage to the outer sheath during the pulling process. Cable clamps and hangers were used to secure the cable neatly inside each manhole.



2. Aerial Cable Installation

Following the completion of the underground cable installation, the team proceeded with the aerial cable installation along the designated utility poles. The following steps were performed during this process:

- A messenger wire was first strung along the utility poles to serve as the main support line for the aerial cable.
- The aerial cable was then lifted and strung from one pole to the next following the planned cable route.
- Cable clamps and hooks were installed on each pole to secure the aerial cable firmly onto the messenger wire.
- The cable was tensioned appropriately to ensure it did not sag excessively between the poles.
- Sufficient cable slack was left at the designated splicing points to allow for the fusion splicing process.



3. Fiber Core Splicing and Signal Verification

Upon reaching the closure location, the fiber splicing process was carried out systematically to ensure each fiber core was properly connected with minimal signal loss. The following steps were performed during this process:

- Each fiber core was individually stripped, cleaned using isopropyl alcohol and lint-free wipes, and carefully loaded into the fusion splicer.
- The fusion splicer automatically aligned and fused each pair of fiber cores together using an electric arc to produce a stable and low-loss joint.
- After each splice was completed, the estimated loss value was displayed on the fusion splicer screen and any reading that exceeded the acceptable threshold was immediately redone to ensure optimal connection quality.
- Once all fiber cores were successfully spliced, an OTDR test was conducted to trace the full optical path and verify that there were no unexpected reflections, breaks, or faults along the fiber route.
- Power meter readings were also taken at the endpoint to verify the end to end signal levels and confirm that the signal transmission met the required performance standards.



4. Fiber Closure Mounting and Cable Termination

Once the fusion splicing process was fully completed, the fiber joint closure was installed to protect all spliced fiber connections. The following steps were carried out during this process:

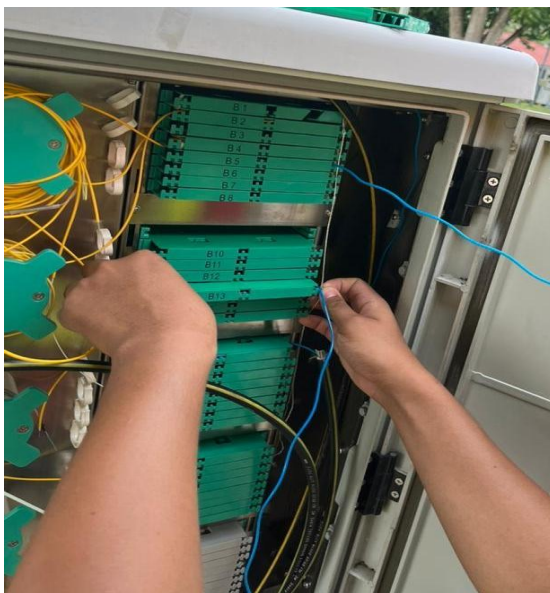
- The fiber joint closure was carefully sealed and mounted onto the designated pole to protect the fiber connections from environmental exposure such as moisture, dust, and physical damage.
- The closure was properly secured and labeled for easy identification during future maintenance work.
- Each fiber core inside the closure was neatly organized and placed into the splicing tray to ensure the fibers were well protected and free from any unnecessary bending or stress.
- A final check was conducted to ensure the closure was fully sealed and all fiber cores were properly secured before the installation was declared complete.



5. FDC Termination and Final Network Connection

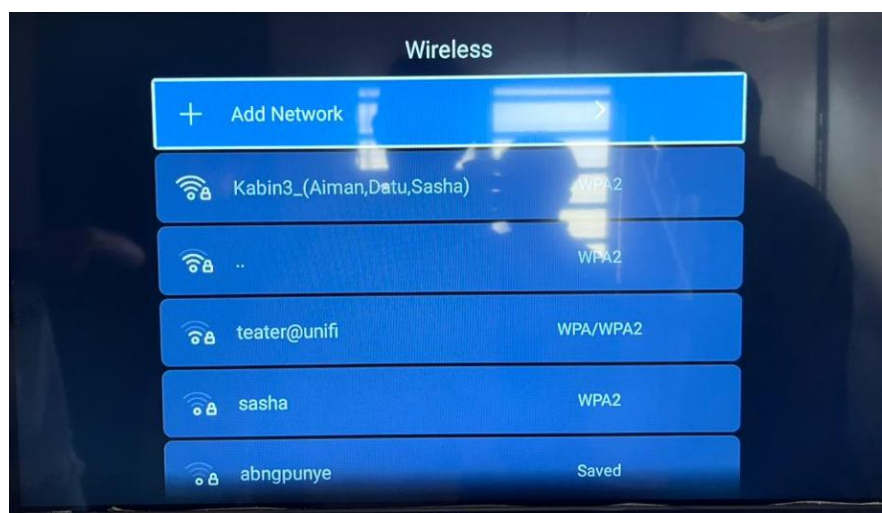
The final stage of the installation involved terminating all fiber cores inside the Fiber Distribution Cabinet (FDC) to complete the network connection. The following steps were carried out during this process:

- All fiber cores were carefully routed and neatly arranged inside the cabinet to ensure a clean and organized termination layout.
- Each fiber core was connected to the designated port on the adapter panel according to the project plan and cable labeling.
- The fiber cables were secured using cable ties and organizers to prevent any unnecessary movement or bending inside the cabinet.
- A final inspection was conducted by the technician to verify that all connections were stable, clean, and functioning properly before the cabinet was closed.
- Once all connections were confirmed to be in good condition, the cabinet was closed and the installation work was officially completed.



6. Media Converter Configuration for Fiber to Wireless Network

The media converter plays an important role in this installation as it was used to convert optical signals transmitted through the fiber optic cable into electrical signals that can be processed by network devices. A pair of media converters operating on a TX/RX (Transmit/Receive) basis were configured and deployed to extend the fiber optic connection from the main cable route and support wireless network access within Cabin 3.



- The TX media converter was connected at the fiber source end via the SC/LC port to transmit the optical signal.
- The RX media converter was placed inside Cabin 3, converting the received optical signal back into an electrical signal through the RJ45 Ethernet port.
- Both media converters were powered on and the LNK/ACT LED indicator was checked to confirm a stable fiber link between both ends.
- A wireless access point (AP) was connected to the RX media converter via RJ45 cable and configured with SSID, WPA2 security, and channel frequency.
- A connectivity test was performed to verify the wireless device successfully obtained an IP address and achieved stable internet access inside Cabin 3.

7. Pon Meter Testing and Signal Level Verification

A PON (Passive Optical Network) meter was used during the final stage of the installation to measure and verify the optical signal power levels at the ONU (Optical Network Unit) end inside Cabin 3. This testing process was carried out to ensure that the fiber optic installation was functioning correctly and that the signal levels received at the endpoint met the required performance standards set for the network.

- The PON meter was connected to the fiber patch cord at the ONU end inside Cabin 3.
- The downstream signal at 1490 nm was measured and compared against the acceptable power range of -8 dBm to -27 dBm.
- The upstream signal at 1310 nm was also tested to verify the ONU was transmitting correctly toward the OLT at the FDC end.
- Any reading outside the acceptable range indicated issues such as splice loss, fiber bending, or dirty connectors that required immediate rectification.
- All readings were within the acceptable threshold, confirming a successful installation with stable optical performance throughout the cable route.



CABLE LENGTH CALCULATION

NO	TYPE OF CABLES	CABLE LENGTH (METER)
1	Underground Cable	12m
2	Aerial Cable	50m
3	Drop Cable	43m
TOTAL		105m

The total fiber optic cable length used in this installation is 105 meters, consisting of three types of cables as follows:

- **Underground Cable** — A total of 12 meters of underground cable was used, routed through the underground duct passing through the designated manholes along the cable path.
- **Aerial Cable** — A total of 50 meters of aerial cable was used, strung along the utility poles from the starting point to the designated splicing location at the closure.
- **Drop Cable** — A total of 43 meters of drop cable was used, routed from the last utility pole down to Cabin 3 as the final connection point to the customer premises.

The measurement was taken based on the actual installation route and field layout using a measurement wheel to ensure accuracy.

CONCLUSION

In conclusion, the fiber optic cable installation project from the Fiber Distribution Cabinet (FDC) to Cabin 3 was successfully completed in accordance with the required technical standards and safety procedures. Throughout the project, every phase was carefully planned and executed, starting from the site survey and cable measurement, followed by the cable installation, fusion splicing, closure mounting, media converter configuration, and final signal verification using a PON meter.

The successful configuration of the TX/RX media converter pair has proven that fiber optic technology can be effectively extended to support wireless network access, providing a stable and reliable Wi-Fi connection within Cabin 3. The PON meter testing conducted at the final stage further confirmed that all optical signal power levels were within the acceptable threshold, ensuring the integrity and performance of the entire fiber optic link from the FDC to the endpoint.

Through this project, students gained valuable hands on experience in handling fiber optic tools and equipment, performing accurate cable measurements, executing proper splicing techniques, configuring media converters, and conducting signal testing using both an OTDR and PON meter. These experiences have significantly enhanced the technical knowledge and practical skills of each team member in the field of fiber optic networking.

Overall, this project serves as a strong foundation for students to develop their competency in modern network infrastructure and prepares them for real world challenges in the telecommunications industry.

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